AI for Communication and Marketing

Mar 18, 2026 - Lab lecture #2: Exploratory Data Analysis for Marketing Applications

What is EDA and Why is it Crucial for Data Science?

Exploratory Data Analysis (EDA)

The systematic process of analyzing datasets to summarize their main characteristics, often using visual methods and statistical summaries.

Why it Matters

Uncovers hidden patterns, anomalies, and relationships within data that drive strategic marketing decisions.

Foundation for Success

Prevents "garbage in, garbage out" by ensuring data quality before building AI models.

EDA Step 1: Dataset Audit – Descriptive Statistics

Objective: Get a Foundational Understanding

Numerical Variables

Calculate mean, median, standard deviation, minimum, maximum, and quartiles to understand distribution and spread.

Categorical Variables

Count frequencies, identify unique values, and assess distribution patterns across categories.

Marketing Application: Understanding customer demographics, purchase history, and campaign engagement metrics to identify high-value segments.

EDA Step 2: Data Quality – The Three Pillars

01

Duplicates

Why they matter: Skew analysis and model training by artificially inflating counts.

Handling: Identify and remove exact duplicate rows using unique identifiers.

02

Missing Values

Why they matter: Can bias results or break algorithms that require complete data.

Handling: Imputation using mean, median, mode, or model-based methods, or strategic removal.

03

Outliers

Why they matter: Can disproportionately influence statistical measures and models.

Handling: Detection using IQR or Z-score methods, followed by capping, transformation, or removal.

EDA Step 3: Creating Key Performance Indicators (KPIs)

Objective: Define Measurable Metrics

Connect data analysis to tangible marketing outcomes that drive business decisions.

Examples:

Avg Spending

Average amount of money spent by each customer

Frequency

Average number of orders made by each customer

Retention Rate

Share of customers that are coming back in a given period of time

The Power of EDA in AI Marketing

EDA is a Mindset

Not just a step, but a data-driven approach to marketing that ensures quality insights.

Essential Foundation

Thorough data auditing, cleaning, and KPI definition are essential for building effective AI models.

Business Alignment

Ensures AI marketing efforts are efficient, impactful, and aligned with strategic objectives.

Next Steps

We will now move to a hands-on session with a sample marketing dataset, applying EDA techniques to uncover actionable insights.

Hands-on session: Your EDA Challenge

customers.csv

Column	Type	Description
`Customer_ID`	string	Unique identifier for each customer
`Age`	integer	Customer age in years at the time of data collection
`Gender`	string	Customer gender (`F` = female, `M` = male)
`Privacy`	string	Whether the customer has given marketing consent (`Y` = yes, `N` = no)
`tenure`	float	Length of the customer relationship in years
`Region`	string	Italian region where the customer is registered

orders.csv

Column	Type	Description
`year`	integer	Calendar year in which the order was placed
`Customer_ID`	string	Unique identifier for each customer
`Order_Nr`	string	Unique order identifier
`Product_category`	string	Product category of the purchased item (`Beauty` or `Electronics`)
`Quantity`	integer	Number of units purchased in the order
`Value`	float	Monetary value of the order in euros

Hands-on Session: Your EDA Challenge

Now, let's put theory into practice. We'll work through a real-world marketing scenario to apply the EDA steps we've discussed.

01

Dataset Audit: Customers & Orders

Begin with a comprehensive audit of `customers.csv` and `orders.csv` to understand their structure and key characteristics.

02

Data Quality: Cleaning Strategy

Identify and address missing values and outliers. Justify your cleaning choices based on feature distributions and potential impact on models.

03

Customer Trends & KPIs

Conduct a descriptive analysis to uncover trends in customer behavior that explain the year-over-year sales decline. Define relevant shopping behavior KPIs.

04

Insights for the CEO

Perform your analysis in a notebook and prepare key findings and actionable recommendations for the CEO, presented as clear slides.